

E-Commerce Sales and Customer Intelligence Dashboard

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2026-10-05

E-Commerce Business Intelligence Dashboard as a real portfolio project.

We'll do it in stages:

1. Define the business problem and requirements
2. Create the e-commerce dataset/database
3. Load data into MySQL
4. Perform advanced SQL analysis
5. Clean/transform data with Power Query
6. Build a proper Star Schema
7. Create DAX measures
8. Build the Power BI dashboard
9. Extract business insights and recommendations

Business scenario

Me as a Data Analyst for an e-commerce company and the management team wants answers to questions such as:

- How much revenue are we generating?
- Which products and categories perform best?
- Which customers generate the most revenue?
- Which regions are performing well?
- Are sales increasing or decreasing?
- What is our profit margin?
- What is the average order value?
- How many customers are returning?
- Which products have poor profitability?
- How is performance changing month-over-month?

Our technology stack

We'll deliberately use the technologies you want to strengthen:

So by the end, I will be able to say I built an end-to-end BI solution starting from relational data in MySQL, performed advanced SQL analysis, transformed the data using Power Query, created a star-schema model, developed DAX measures and built an interactive Power BI dashboard.

Business requirements

this is an important part that most student Power BI projects skip.

We'll create a small Business Requirements Document.

Stakeholders

E-Commerce Management

Business objective:

Develop an interactive business intelligence dashboard that enables management to monitor sales, profitability, customer behavior and product performance.

Key business questions:

- Sales
 1. What is total revenue?
 2. What is total profit?
 3. What is the monthly sales trend?
 4. What is the month-over-month growth?
- Customers
 1. Who are our highest-value customers?
 2. What percentage of customers are returning?
 3. Which customer segments generate the most revenue?
- Products
 1. Which products generate the most revenue?
 2. Which products generate the most profit?
 3. Which categories have poor margins?
- Geography
 1. Which regions generate the most revenue?
 2. Which stores perform best?
- Management
 1. What are our key KPIs?
 2. Where are the biggest opportunities for improvement?